AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M140
Introducing Statistics
Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19 /Nov./2025

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed: 120 minutes**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions:4
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 30% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

CamScanner

Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 3 marks, giving a total of 12 possible marks.

Q–1: The total area under the standard normal curve is

a) 0
b) 0.5
c) 1
d) 2

Q–2: The mode of the sample data 29, 27, 23, 29, 32, 30, 23, 29, 26 is

a) 32
b) 23
c) 26
d) 29

Q–3: If two events are independent, we can _____ their probabilities to determine the intersection probability.

a) Divide
b) Multiply
c) Subtract
d) Add

Q–4: Which percentile describes Q2:

a) 25th
b) 50th
c) 75th
d) 100th

Part 2: ESSAY QUESTIONS

Each question is worth 16 marks. You should answer all the three questions, giving a total of 48 possible marks.

Q–1: [4+4+8 Marks]

The following sample data represents the diastolic blood pressure (in mm Hg.) of 11 persons:

71 81 85 94 88 84 82 86 86 79 77

a) Compute the sample mean.
b) Compute the sample median.
c) Find the sample standard deviation.

Q-2: [16 Marks]

Find the value of the correlation coefficient from the following table:

Age	x	43	21	25	42	57	59
Glucose Level	y	99	65	79	75	87	81

CamScanner

Q–3: [4*4 =16 Marks]

AOU has to decide on a new approach to teaching Statistics. Full-time and part-time tutors were surveyed and the results are summarized below.

	Prefer Lecture (Tutorial)	Prefer Computer (Online)	No Preference	Total
Full-time	10	20	25	55
Part-time	12	25	8	45
Total	22	45	33	100

a) If a tutor is selected at random, what is the probability that he is a part-timer?
b) If a tutor is selected at random, what is the probability that he has no preferences?
c) If a tutor is selected at random, what is the probability that he is a part-timer and has no preferences?
d) If a tutor is selected at random, what is the probably that he is a part-timer GIVEN he Prefer Computer (Online)?

End of Questions

AOU EGYPT

CamScanner